FINAL YEAR MDM PROJECT · SUPPLY CHAIN ANALYTICS

DATA SCIENCE AND ANALYTICS FOR FINTECH - 7MD470

Supplier Selection Scorecard

A data-driven model that ranks suppliers on Cost Efficiency, Delivery Reliability and Supply Chain Resilience



Cost



Reliability



Resilience

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The Problem: Price Alone Picks the Wrong Supplier

Which supplier should this order be sourced from?

Problem statement

Given historical order, shipment, cost and delivery data, build a model that scores and ranks suppliers on Cost Efficiency, Delivery Reliability and Resilience, and combines them into one composite scorecard for sourcing decisions.

Why it matters

- Procurement teams manage 50–500+ suppliers — manual comparison does not scale.
- A cheap supplier that delivers late causes stock-outs, expedited freight and lost sales.
- Post-COVID and Suez-blockage disruptions show that optimising on unit cost alone raises risk.

What we answer



Who is cheapest like-for-like?
Price vs. same-product benchmark + freight



Who delivers on time?
On-time rate and average delay



Who stays consistent?
Delay spread and month-to-month stability



How robust is the ranking?
Sensitivity to pillar weights

The Intuition: A Football Transfer Analogy

Same multi-criteria decision pattern, different domain



Football transfer model	Supplier scorecard model
Player	Supplier
Transfer fee / wages	Unit cost / total landed cost
Profitability (return for the club)	Cost Efficiency — price vs. market benchmark
Popularity (fan trust)	Delivery Reliability — on-time rate, the "trust proxy"
Win probability (helps team win)	Resilience — consistency under disruption
Composite "transfer value"	Composite supplier score
Recommend which player to sign	Recommend which supplier to source from

Just as a club would not sign the cheapest player regardless of form, a buyer should not pick the cheapest supplier regardless of delivery record.

Dataset & Data Cleaning

USAID Supply Chain Shipment Pricing Data (SCMS Delivery History, via Kaggle)

10,324
shipment records

33
columns

73
unique vendors

25
vendors ranked (≥ 20 shipments)

4,126
non-numeric freight values handled

2006–15
scheduled delivery period

Why this dataset: it has a real, named Vendor field, plus scheduled vs. delivered dates, unit price, product group, freight cost and line-item value.

Cleaning pipeline

1. Read CSV (UTF-8 or Windows-1252)
2. Drop rows with no vendor or bad dates
3. Parse dates: 2-Jun-06, 2006-06-02, 06/02/2006
4. Text in numeric cells → missing, not 0
5. Delay = Delivered − Scheduled (days)
6. Exclude vendors with < 20 shipments

System Workflow

A browser-based app: upload any shipment CSV and get a ranked scorecard in five steps



01

Upload

Any order / shipment CSV



02

Map columns

Auto-detects vendor, dates, price, freight



03

Check data

Rows dropped, bad values, eligible vendors



04

Set weights

Sliders or presets + min. shipments



05

Results

Scorecard, sensitivity, baseline, CSV export



Architecture: 100% client-side

No server, no install — open index.html in Chrome, Firefox or Edge. Data never leaves the user's machine.

index.html
page structure

style.css
visual design

script.js
cleaning, scoring, rendering

PapaParse
CSV parsing

Methodology: From Raw Shipments to One Score

Six engineered metrics → three pillars → one composite

Cost Efficiency

Price index
avg unit price ÷ median price of same product group

Freight ratio
Σ freight cost ÷ Σ line-item value

Delivery Reliability

On-time rate
share of shipments with delay ≤ 0

Average delay
mean of (delivered − scheduled) days

Resilience

Delay spread
standard deviation of delay

Monthly consistency
std. dev. of monthly on-time rate (months with ≥ 3 shipments)

Normalise
Min-max scale each metric to 0–100; invert "lower is better" metrics

→

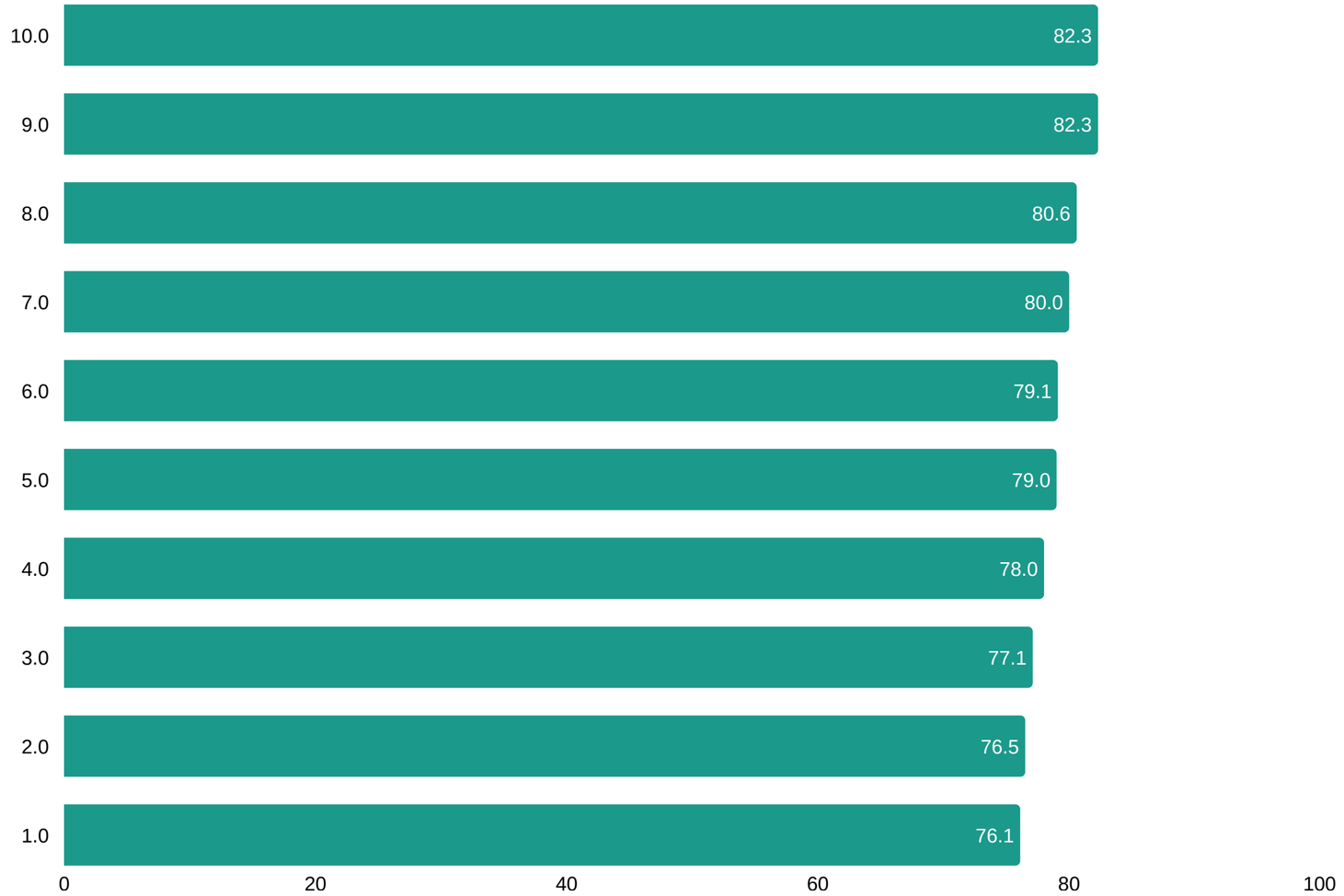
Pillar score
Mean of the pillar's available normalised metrics

→

Composite
$\Sigma (w_p \times \text{Pillar}_p) \div \Sigma w_p$ — weights set by user

Results: Top 10 Suppliers (Equal Weights)

25 vendors ranked · 88.5% of all shipments arrived on or before the scheduled date



Key finding: cheap ≠ best

Vendor	Price rank	Score rank
Aurobindo Pharma	5	**22**
Cipla	6	**24**
Micro Labs	1	**8**

Aurobindo and Cipla are among the cheapest vendors, but only ~86–87% of their shipments are on time and they average 4–5 days late — so they fall to the bottom of the scorecard. A price-only ranking would have recommended them.

Sensitivity Analysis: Does the Ranking Hold Up?

Rank of today's top suppliers under four weighting scenarios (Cost : Reliability : Resilience)

Supplier	Equal 1:1:1	Cost 3:1:1	Reliability 1:3:1	Resilience 1:1:3
Bristol-Myers Squibb	**1**	**1**	**2**	**1**
Pharmacy Direct	**2**	**2**	**3**	**2**
Hoffmann-La Roche	**3**	6	4	**3**
GlaxoSmithKline Export	4	8	6	4
Mylan Laboratories	5	**3**	7	8
Hetero Labs	6	7	8	6
Aspen Pharmacare	7	5	13	7
Micro Labs	8	15	9	5

Top 3 Drops out of top 8

Robust leaders

Bristol-Myers Squibb and Pharmacy Direct stay in the top 3 under every weighting.

Hidden cost

Micro Labs falls from 8th to 15th when cost dominates — its freight overhead is high despite a low unit price.

Why it matters

A recommendation that holds across scenarios is safer to act on than one fixed score.

Conclusion & Future Scope

What we delivered

- Working web app that turns any shipment CSV into a ranked, explainable supplier scorecard
- Robust cleaning of messy real-world data (encodings, date formats, text in numeric columns)
- Six metrics across three pillars, user-adjustable weights and a min-shipment filter
- Sensitivity table and price-only baseline show why multi-criteria beats price alone

Project Repository: DS-A_Project
(Click on project name to view project repo)

Limitations → next steps

- AHP / TOPSIS weighting instead of manual weights
- Optimisation layer: allocate volume across suppliers (linear programming with PuLP)
- Backtest the scorecard on a held-out time period
- Richer resilience: seasonal peaks, shipment mode, recovery after shocks
- Test generalisation on the DataCo retail dataset

Current resilience metrics are proxies (delay spread, monthly consistency), not full disruption analysis.